

CO4 – AT3 COMPARATIVE ANALYSIS

Answered Questions

The answers below follow the questions, values, and requested calculations in the uploaded Comparative Analysis task.

1. Object Detection Performance Comparison

Question: Two object detection models, Viola–Jones and YOLO, produce the following results: Model A has 80 True Positives, 20 False Positives, and 30 False Negatives, while Model B has 90 True Positives, 15 False Positives, and 20 False Negatives. (a) Calculate the precision and recall for both models. (b) Compare the results and determine which model provides better overall object detection performance. Justify your answer based on the calculated values.

Answer:

Model	TP	FP	FN	Precision	Recall
Model A (Viola–Jones)	80	20	30	$80/(80+20) = 80.00\%$	$80/(80+30) = 72.73\%$
Model B (YOLO)	90	15	20	$90/(90+15) = 85.71\%$	$90/(90+20) = 81.82\%$

Precision formula: Precision = TP / (TP + FP).

For Model A: $80 / (80 + 20) = 80 / 100 = \mathbf{80.00\%}$.

For Model B: $90 / (90 + 15) = 90 / 105 = \mathbf{85.71\%}$.

Recall formula: Recall = TP / (TP + FN).

For Model A: $80 / (80 + 30) = 80 / 110 = \mathbf{72.73\%}$.

For Model B: $90 / (90 + 20) = 90 / 110 = \mathbf{81.82\%}$.

Comparison: Model B performs better on both measures. Its precision is 85.71%, compared with 80.00% for Model A, so Model B produces fewer false-positive detections relative to its total predicted positives. Its recall is 81.82%, compared with 72.73% for Model A, so Model B detects a larger proportion of the actual positive objects.

Final decision: Model B (YOLO). Based on the calculated precision and recall, Model B provides better overall object detection performance because it has both higher precision and higher recall than Model A.

2. Motion Estimation Error Analysis

Question: Two motion estimation techniques produce the following displacement errors in pixels: Method A: 2, 3, 2 and Method B: 3, 5, 4 for three consecutive frames. (a) Calculate the average displacement error for both methods. (b) Compare the two methods and determine which method provides more accurate motion estimation. (c) Justify your answer based on the error values.

Answer:

Method	Frame 1	Frame 2	Frame 3	Total Error	Average Error
Method A	2	3	2	$2+3+2 = 7$	$7/3 = 2.33$ pixels
Method B	3	5	4	$3+5+4 = 12$	$12/3 = 4.00$ pixels

Method A average displacement error: $(2 + 3 + 2) / 3 = 7 / 3 = \mathbf{2.33}$ pixels.

Method B average displacement error: $(3 + 5 + 4) / 3 = 12 / 3 = \mathbf{4.00}$ pixels.

Comparison: Method A has the lower average displacement error, 2.33 pixels, whereas Method B has an average error of 4.00 pixels. Since lower displacement error means that the estimated motion is closer to the expected motion, Method A provides more accurate motion estimation.

Final decision: Method A. Method A is more accurate because its average error is 1.67 pixels lower than Method B ($4.00 - 2.33 \approx 1.67$ pixels). Its individual errors are also lower for all three consecutive frames.

3. Depth Estimation Comparison

Question: A stereo vision system and a Structure-from-Motion (SfM) system estimate the depth of three objects. The estimated depths are: Stereo Vision: 2.0 m, 3.0 m, 4.0 m and SfM: 2.5 m, 3.8 m, 5.0 m, while the actual depths are 2 m, 3 m, and 4 m, respectively. (a) Calculate the absolute error for each object for both methods. (b) Calculate the average absolute error for both methods. (c) Compare the methods and identify which provides more accurate depth estimation.

Answer: Absolute error = $|\text{Estimated depth} - \text{Actual depth}|$.

Object	Actual Depth	Stereo Estimate	Stereo Absolute Error	SfM Estimate	SfM Absolute Error
1	2.0 m	2.0 m	$ 2.0 - 2.0 = 0.0$ m	2.5 m	$ 2.5 - 2.0 = 0.5$ m
2	3.0 m	3.0 m	$ 3.0 - 3.0 = 0.0$ m	3.8 m	$ 3.8 - 3.0 = 0.8$ m
3	4.0 m	4.0 m	$ 4.0 - 4.0 = 0.0$ m	5.0 m	$ 5.0 - 4.0 = 1.0$ m

Stereo Vision average absolute error: $(0.0 + 0.0 + 0.0) / 3 = \mathbf{0.00}$ m.

SfM average absolute error: $(0.5 + 0.8 + 1.0) / 3 = 2.3 / 3 = \mathbf{0.77}$ m approximately.

Comparison: Stereo Vision has zero absolute error for all three objects, giving an average absolute error of 0.00 m. SfM has errors of 0.5 m, 0.8 m and 1.0 m, with an average of approximately 0.77 m. Therefore, Stereo Vision provides the more accurate depth estimation for the data given in the question.

Final decision: Stereo Vision. Its estimates exactly match the actual depths for all three objects, whereas SfM overestimates the depth of every object.

Overall Conclusion

Based strictly on the supplied numerical data: Model B (YOLO) gives better object detection performance because it has higher precision and recall; Method A gives more accurate motion estimation because it has the lower average displacement error; and Stereo Vision gives more accurate depth estimation because its average absolute error is 0.00 m.